

## ALMERIA AP, Spain

WMO: 084870

Lat: 36.846N

Lon: 2.357W

Elev: 21

StdP: 101.07

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 6.5        | 7.6 | -6.8                          | 2.1 | 11.4 | -4.1 | 2.7 | 12.5 | 15.2                  | 15.3 | 13.4 | 14.7 | 4.5                      | 340  | 0.414 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 7.2                    | 34.3            | 21.0 | 32.7 | 21.1 | 31.1 | 21.3 | 25.6                | 28.7 | 24.8 | 28.3 | 24.1 | 27.8 | 8.6                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.7                            | 19.8 | 27.7 | 23.7 | 18.6 | 27.2 | 22.9 | 17.7 | 26.7 | 78.6          | 28.6 | 75.2 | 28.6 | 72.5 | 27.9 | 29.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 13.0              | 11.5 | 10.2 | DB                         | 4.5 | 37.8               | 1.7 | 1.5                                                | 3.3  | 38.9       | 2.3  | 39.7       | 1.4  | 40.6       | 0.2  | 41.6 |  |
|                   |      |      | WB                         | 0.6 | 26.3               | 1.9 | 1.6                                                | -0.8 | 27.4       | -1.9 | 28.3       | -3.0 | 29.2       | -4.4 | 30.4 |  |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 19.2   | 12.8  | 13.3  | 15.0  | 17.0  | 19.9  | 23.5  | 26.1  | 26.9  | 24.1  | 20.8  | 16.2  | 13.8 |      |
|                                                                                  | DBStd                                                                            | 5.37   | 1.77  | 1.95  | 1.89  | 2.09  | 2.41  | 2.70  | 2.46  | 2.18  | 2.22  | 2.19  | 2.29  | 1.83 |      |
|                                                                                  | HDD10.0                                                                          | 5      | 2     | 2     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |      |
|                                                                                  | HDD18.3                                                                          | 694    | 172   | 141   | 104   | 52    | 10    | 0     | 0     | 0     | 4     | 70    | 141   |      |      |
|                                                                                  | CDD10.0                                                                          | 3344   | 88    | 94    | 156   | 209   | 307   | 406   | 499   | 523   | 423   | 333   | 187   | 118  |      |
|                                                                                  | CDD18.3                                                                          | 992    | 0     | 0     | 1     | 11    | 59    | 156   | 241   | 264   | 173   | 79    | 7     | 0    |      |
|                                                                                  | CDH23.3                                                                          | 7365   | 0     | 0     | 4     | 45    | 250   | 999   | 2138  | 2568  | 1106  | 247   | 7     | 1    |      |
|                                                                                  | CDH26.7                                                                          | 2284   | 0     | 0     | 0     | 4     | 59    | 323   | 722   | 888   | 261   | 28    | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 4.0    | 3.9   | 4.2   | 4.3   | 4.5   | 4.2   | 4.0   | 3.7   | 3.8   | 3.9   | 3.7   | 3.9   | 3.9  |      |
| Precipitation                                                                    | PrecAvg                                                                          | 200    | 23    | 20    | 19    | 21    | 15    | 6     | 1     | 1     | 10    | 30    | 27    | 28   |      |
|                                                                                  | PrecMax                                                                          | 553    | 112   | 75    | 69    | 102   | 69    | 85    | 9     | 6     | 107   | 104   | 101   | 103  |      |
|                                                                                  | PrecMin                                                                          | 64     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                                  | PrecStd                                                                          | 87     | 25    | 19    | 18    | 21    | 18    | 14    | 2     | 19    | 27    | 26    | 27    |      |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 19.9  | 20.9  | 23.8  | 27.2  | 31.0  | 35.0  | 36.6  | 36.5  | 32.7  | 29.2  | 24.4  | 21.1 |      |
|                                                                                  |                                                                                  | MCWB   | 13.2  | 13.6  | 13.9  | 15.8  | 17.5  | 19.9  | 20.5  | 21.1  | 21.5  | 19.4  | 17.0  | 14.7 |      |
|                                                                                  | 2%                                                                               | DB     | 18.1  | 19.1  | 21.2  | 24.6  | 28.2  | 32.5  | 34.4  | 34.2  | 31.0  | 27.1  | 22.5  | 19.3 |      |
|                                                                                  |                                                                                  | MCWB   | 13.3  | 13.2  | 13.6  | 15.0  | 16.6  | 19.1  | 20.5  | 21.9  | 21.0  | 19.6  | 16.8  | 14.5 |      |
|                                                                                  | 5%                                                                               | DB     | 17.2  | 18.0  | 19.9  | 22.2  | 26.1  | 30.2  | 32.4  | 32.7  | 29.4  | 25.7  | 21.2  | 18.3 |      |
|                                                                                  |                                                                                  | MCWB   | 12.9  | 13.2  | 13.7  | 14.6  | 16.5  | 18.9  | 21.2  | 22.0  | 21.6  | 19.7  | 16.3  | 14.1 |      |
|                                                                                  | 10%                                                                              | DB     | 16.4  | 17.1  | 18.8  | 20.7  | 24.1  | 28.1  | 30.3  | 31.0  | 28.0  | 24.6  | 20.1  | 17.4 |      |
|                                                                                  |                                                                                  | MCWB   | 12.4  | 12.9  | 13.7  | 14.9  | 16.8  | 19.5  | 21.4  | 22.5  | 21.7  | 19.5  | 15.7  | 13.6 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 15.8  | 16.1  | 17.1  | 18.1  | 21.3  | 24.4  | 26.3  | 27.1  | 25.7  | 22.7  | 19.7 | 17.1 |
|                                                                                  |                                                                                  |        | MCDB  | 17.6  | 17.9  | 19.4  | 21.6  | 24.2  | 27.5  | 29.3  | 29.7  | 28.0  | 25.1  | 21.7 | 19.1 |
| 2%                                                                               |                                                                                  | WB     | 14.8  | 15.1  | 16.0  | 17.2  | 19.9  | 23.0  | 25.2  | 25.9  | 24.5  | 21.7  | 18.4  | 16.0 |      |
|                                                                                  |                                                                                  | MCDB   | 17.0  | 17.4  | 18.6  | 20.6  | 23.1  | 26.3  | 28.7  | 28.9  | 27.6  | 24.9  | 21.0  | 17.9 |      |
| 5%                                                                               |                                                                                  | WB     | 13.9  | 14.3  | 15.2  | 16.5  | 19.1  | 22.1  | 24.3  | 25.1  | 23.6  | 20.9  | 17.3  | 15.1 |      |
|                                                                                  |                                                                                  | MCDB   | 16.4  | 16.9  | 18.1  | 20.0  | 22.7  | 26.0  | 28.2  | 28.6  | 27.0  | 24.2  | 20.1  | 17.4 |      |
| 10%                                                                              |                                                                                  | WB     | 13.1  | 13.6  | 14.6  | 15.8  | 18.3  | 21.3  | 23.6  | 24.4  | 22.7  | 20.2  | 16.4  | 14.3 |      |
|                                                                                  |                                                                                  | MCDB   | 15.7  | 16.3  | 17.6  | 19.4  | 22.1  | 25.2  | 27.5  | 28.2  | 26.4  | 23.6  | 19.3  | 16.7 |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 7.5   | 7.4   | 7.5   | 7.4   | 7.4   | 7.6   | 7.3   | 7.2   | 7.2   | 7.0   | 7.1  | 7.4  |
|                                                                                  |                                                                                  |        | MCDBR | 8.1   | 8.3   | 8.9   | 9.6   | 10.4  | 11.1  | 10.6  | 10.0  | 8.7   | 8.2   | 7.7  | 8.0  |
|                                                                                  | MCWBR                                                                            |        | 6.4   | 6.0   | 5.6   | 5.0   | 5.1   | 5.0   | 5.1   | 5.2   | 4.8   | 5.1   | 5.5   | 6.1  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 7.8   | 7.7   | 7.8   | 7.7   | 7.4   | 7.3   | 7.1   | 7.1   | 7.0   | 7.1   | 7.0   | 7.4  |      |
|                                                                                  |                                                                                  | MCWBR  | 6.8   | 6.5   | 6.1   | 5.3   | 5.1   | 5.1   | 5.6   | 5.5   | 5.7   | 5.7   | 5.8   | 6.4  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.322  | 0.351 | 0.395 | 0.397 | 0.421 | 0.449 | 0.461 | 0.462 | 0.429 | 0.399 | 0.354 | 0.329 |      |      |
|                                                                                  | taud                                                                             | 2.516  | 2.450 | 2.311 | 2.340 | 2.299 | 2.231 | 2.194 | 2.202 | 2.317 | 2.393 | 2.487 | 2.516 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 871    | 882   | 867   | 885   | 865   | 837   | 824   | 814   | 824   | 816   | 823   | 836   |      |      |
|                                                                                  | Edh at Noon                                                                      | 80     | 96    | 121   | 124   | 131   | 140   | 144   | 140   | 119   | 101   | 82    | 76    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 2.71   | 3.59  | 4.83  | 5.96  | 6.94  | 7.73  | 7.61  | 6.71  | 5.32  | 3.94  | 2.85  | 2.40  |      |      |
|                                                                                  | RadStd                                                                           | 0.22   | 0.35  | 0.38  | 0.41  | 0.49  | 0.22  | 0.15  | 0.23  | 0.27  | 0.26  | 0.23  | 0.18  |      |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +72     |
| (47) | Regional (5 neighbors) | +0.32 | N/A     | N/A    | +0.79   | +0.30 | N/A   | N/A         | N/A     | +152    | +105    |

Nomenclature: See separate page